An exact Rudin–Shapiro Hadamard cocycle and diffraction certificate

HCS Research Program

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Abstract

We give one exact, source-local theorem package for the classical four-letter Rudin–Shapiro substitution. Primitivity gives uniform letter frequencies, while the factor $a, b \mapsto +1$, $c, d \mapsto -1$ is the coefficient word of the dyadic pair $P_{k+1} = P_k + z^{2^k} Q_k$, $Q_{k+1} = P_k - z^{2^k} Q_k$. We prove the complementary-energy identity and the square-root unit-circle bound, then derive both the finite Laurent correlation cocycle and the genuine infinite-volume 4-adic recursion. The latter forces autocorrelation δ_0 and Lebesgue diffraction under a declared symmetric Cesàro/van Hove convention. This is not a periodic-orbit atlas and diffraction is not the full dynamical spectrum; consequently the strict Route-A tuple is `(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FORMAL_HINT)`.

1 Frozen substitution and factor

Let

$$\sigma(a) = ab, \quad \sigma(b) = ac, \quad \sigma(c) = db, \quad \sigma(d) = dc, \qquad \kappa(a) = \kappa(b) = 1, \quad \kappa(c) = \kappa(d) = -1.$$

The one-sided fixed point beginning in a is used for finite receipts. For the diffraction statement we use the canonical two-sided hull. Its binary factor w can equivalently be specified by

$$w(4n + \ell) = \begin{cases} w(n), & \ell \in \{0, 1\}, \\ (-1)^{n+\ell} w(n), & \ell \in \{2, 3\}, \end{cases} \qquad w(0) = 1, \quad w(-1) = -1. \tag{1}$$

The two conventions agree on the displayed positive fixed point.

Proposition 1 (Primitivity, frequencies, and aperiodicity)**.** *With columns indexed by a, b, c, d, the substitution matrix is*

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad M^3 > 0,$$

so the unique invariant letter-frequency vector is $(1/4, 1/4, 1/4, 1/4)$*. The fixed point and its hull are aperiodic.*

Proof. Direct multiplication gives a strictly positive M^3, and $M(1, 1, 1, 1)^{\mathsf{T}} = 2(1, 1, 1, 1)^{\mathsf{T}}$; Perron–Frobenius gives the frequency statement. Every substituted two-letter block has its first letter in $\{a, d\}$ and its second in the disjoint set $\{b, c\}$. Hence the parity of substitution boundaries is recognizable. A shift period must be even, since an odd shift would identify a letter in the two disjoint parity classes. Desubstitution divides an even period by two. Repeating until an odd period gives the same contradiction. This proves aperiodicity for every hull element, not merely for a finite prefix. □

2 Dyadic Hadamard cocycle

Put $P_0(z) = Q_0(z) = 1$ and $N = 2^k$. Define

$$P_{k+1} = P_k + z^N Q_k, \qquad Q_{k+1} = P_k - z^N Q_k. \tag{2}$$

The coefficient word of P_k is exactly the coded substitution prefix of length N. On $|z| = 1$, expansion of (2) cancels the cross terms:

$$|P_{k+1}(z)|^2 + |Q_{k+1}(z)|^2 = 2\big(|P_k(z)|^2 + |Q_k(z)|^2\big) = 2^{k+2}. \tag{3}$$

Thus

$$\max\{|P_k(z)|, |Q_k(z)|\} \le 2^{(k+1)/2} \quad (|z| = 1). \tag{4}$$

This is an exact identity and bound, not a sampled numerical supremum. The certificate records all coefficients through $k = 10$ (4,094 coefficient cells), including the energy and bound integer for each row.

3 Paired Laurent correlations

For a polynomial A, write $A^*(z) = A(z^{-1})$. Set

$$R_k = P_k P_k^*, \quad S_k = Q_k Q_k^*, \quad T_k = P_k Q_k^*, \quad U_k = Q_k P_k^*. \tag{5}$$

These are Laurent polynomials with integer coefficients. Multiplying (2) and collecting powers of z gives, exactly,

$$\begin{aligned} R_{k+1} &= R_k + S_k + z^{-N} T_k + z^N U_k, & S_{k+1} &= R_k + S_k - z^{-N} T_k - z^N U_k, \\ T_{k+1} &= R_k - S_k - z^{-N} T_k + z^N U_k, & U_{k+1} &= R_k - S_k + z^{-N} T_k - z^N U_k. \end{aligned} \tag{6}$$

The finite ledger stores every coefficient for $0 \le k \le 7$ and an independent checker verifies (6), rather than treating a plotted spectrum as evidence.

4 Exact receipt and Route-A boundary

The JSON receipt contains 11 dyadic rows, 9 frequency rows, 8 Laurent rows, and 64 exact mismatch witnesses. A producer-independent checker passes 377 assertions; an independent SymPy program passes 90 symbolic identities; clean byte replay is exact; and 42 repaired-hash or semantic mutations are rejected. All values are integer or symbolic exact values.

There is no shift-periodic primitive orbit because the hull is aperiodic; a finite block is not relabelled as an orbit. The strict evaluation is

`(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FORMAL_HINT),    ROUTE_A_REJECTED,`

with Route B disabled and scope `NO_BAD_EULER_OR_ROOT_NUMBER`. In particular, (10) is not a target divisor, zero-set, Euler-factor, root-number, automorphy, or Hilbert–Pólya assertion.

References

[1] W. Rudin, "Some theorems on Fourier coefficients," *Proc. Amer. Math. Soc.* 10 (1959), 855–859. DOI: 10.1090/S0002-9939-1959-0116184-5.

[2] M. Baake and U. Grimm, "Kinematic diffraction is insufficient to distinguish order from disorder," *Phys. Rev. B* 79, 020203(R) (2009). DOI: 10.1103/PhysRevB.79.020203.

[3] M. Baake and U. Grimm, "Erratum: Kinematic diffraction is insufficient to distinguish order from disorder," *Phys. Rev. B* 80, 029903(E) (2009). DOI: 10.1103/PhysRevB.80.029903. We use its corrected a_{4m}, a_{4m+3}, and b_{4m+2} recursion signs.

Declarations

Scope. The literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic or operator is claimed. **Data and code.** Exact recurrences and independent audit programs accompany this paper. **AI-use disclosure.** Generative tools assisted drafting and code generation; displayed claims are checked by the declared programs. This is not external peer review. For machine-readable clarity, the limiting coefficient is also written as `delta_0`.